

VoiceVerse

Sample Data, Test Cases and Originality Check

Project: AI-powered multilingual content localization for vocational training (LinguaSkill / VoiceVerse), Innovative Design Lab-I. Prepared from the IDL-I pitch deck and the revised IEEE paper supplied by the team.

Item	Count	Where
Vocational glossary entries	16	Section 2.1
Source test sentences	16	Section 2.2
ASR audio scenarios (spec only)	10	Section 2.3
Upload file cases	12	Section 2.4
Test cases	53	Section 3
Worked metric examples (computed by script)	4	Section 4
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1. Scope and how to use this pack

The pack covers the ten-stage pipeline in the paper (upload, Whisper ASR, spaCy preprocessing, FAISS retrieval, IndicTrans2 translation, LangChain localization, Coqui TTS, BLEU evaluation) plus the React/FastAPI layers. Every sentence, glossary row and test case was written fresh for this report; none is copied from the paper or the deck.

Important limits of the sample data. (a) No Indic font is available in the PDF toolchain used here, so Hindi and Tamil renderings are given in Roman script and are **illustrative only**. Where a rendering is not given, it is marked 'expert to confirm'. Your bilingual domain experts must create the real reference translations before any BLEU or terminology score is computed. (b) Audio items are specifications, not recordings; record or synthesize them to match. (c) Pass/fail bounds for ASR are taken from the paper's reported WER values so tests check your own claims.

2. Sample data

2.1 Vocational glossary (retrieval and terminology tests)

ID	English term	Domain	Hindi (Roman, illustrative)	Tamil (Roman, illustrative)	Typical generic-MT error
G01	fuse	Electrical	fyooz (keep as loanword)	fyooz	Rendered as 'fuze' (detonator) or 'melt'
G02	earthing	Electrical	arthing / bhu-sampark	expert to confirm	'Earth' rendered as soil or planet
G03	insulation	Electrical	rodhan / kuchalak	expert to confirm	Rendered as 'isolation' (social sense)
G04	voltage	Electrical	voltata / vidyut dabav	expert to confirm	Rendered as 'pressure' only
G05	torque wrench	Automotive	tark renj	expert to confirm	'Torque' rendered as 'twist/rotation'
G06	brake pad	Automotive	brek pad	expert to confirm	Rendered as 'brake cushion/notepad'
G07	dipstick	Automotive	tel-mapak chhad (oil gauge rod)	expert to confirm	Word-by-word: 'dip' + 'stick'
G08	drip irrigation	Agriculture	boond-boond sinchai	thuli neerpaasanam	Rendered as 'dripping watering'
G09	fertilizer	Agriculture	urvarak	uram	Confused with 'manure' only
G10	pesticide	Agriculture	keetnashak	poochi kolli	Rendered as 'insect medicine'
G11	autoclave	Healthcare	autoklev	expert to confirm	Transliterated wrongly / dropped
G12	blood pressure	Healthcare	rakt chaap	ratha azhuththam	'Pressure' as 'stress'
G13	first aid	Healthcare	prathmik upchar	mudhal uthavi	Rendered as 'first help'
G14	scaffolding	Construction	expert to confirm	expert to confirm	Rendered as 'platform for executions'

ID	English term	Domain	Hindi (Roman, illustrative)	Tamil (Roman, illustrative)	Typical generic-MT error
G15	pipe wrench	Plumbing	paip renj	expert to confirm	'Wrench' rendered as 'twist'
G16	hand hygiene	Healthcare	haath ki safai	expert to confirm	Generic 'hand cleanliness' (loses protocol sense)

2.2 Source sentences (translation, localization and TTS tests)

ID	Domain	English source	Gloss. ref	What it tests
S01	Electrical	Switch off the main supply before replacing the fuse.	G01	Term disambiguation (fuse); imperative preserved
S02	Electrical	Check that the earthing wire is connected to the earth pit.	G02	'earth' appears twice with electrical sense
S03	Automotive	Apply 25 N-m of torque to the wheel nuts using a torque wrench.	G05	Number+unit preserved verbatim
S04	Automotive	Use 10W-40 engine oil and do not fill past the MAX mark on the dipstick.	G07	Alphanumeric spec + negation preserved
S05	Agriculture	Drip irrigation applies water directly to the root zone of each plant.	G08	Compound term must not be literal
S06	Agriculture	Wear gloves when handling pesticide and keep it away from children.	G10	Safety instruction; two clauses
S07	Agriculture	Apply 50 kg of urea fertilizer per acre in two equal doses.	G09	Unit 'acre' / 'kg' localization policy
S08	Healthcare	Follow hand hygiene steps: wash your hands for at least 20 seconds before touching the patient.	G16	Numeric threshold 'at least 20'
S09	Healthcare	Sterilize the instruments in an autoclave at 121 °C for 15 minutes.	G11	Temperature and time preserved
S10	Electrical	Do not touch the live wire with wet hands, even if the insulation looks intact.	G03	Negation is safety-critical; 'live' = energized
S11	Healthcare	Measure the blood pressure after the patient has rested for five minutes.	G12	Number word; 'pressure' sense
S12	Plumbing	Tighten the joint with a pipe wrench, then check every joint for leaks.	G15	Repeated term 'joint'
S13	Construction	Mix cement and sand in a 1:4 ratio before adding water.	-	Ratio '1:4' must survive tokenization
S14	Construction	Scaffolding must be inspected by a supervisor before each shift.	G14	Modal 'must' (obligation) preserved
S15	Automotive	Replace the brake pad if its thickness is below 3 mm.	G06	Threshold direction (below) preserved
S16	Electrical	Part No. AX-4471 is rated for a voltage of 230 V and a current of 16 A only.	G04	Part number must NOT be translated

2.3 ASR scenarios (metadata only)

ID	Scenario	Duration	Speech	Noise	Accent	WER bound
A01	Short safety briefing	1 min 30 s	English	Quiet room	Neutral	<= 5%
A02	Electrical wiring lecture	12 min	English	Classroom hum	Neutral	<= 7%
A03	Workshop demonstration	8 min	English	Machine noise (~65 dB, simulated)	Neutral	<= 10%
A04	Long theory lecture	28 min	English	Quiet room	Neutral	<= 8%
A05	Technical vocabulary dictation (G01-G16)	4 min	English	Quiet room	Neutral	<= 9.5%
A06	Tamil-accented English lecture	10 min	English	Quiet room	Tamil-accented	<= 11.5%
A07	Hindi-accented English lecture	10 min	English	Quiet room	Hindi-accented	<= 11.5%
A08	Code-mixed Hindi-English instruction	3 min	Hinglish	Quiet room	Mixed	Report only
A09	Silence only (negative)	30 s	None	Silence	-	Empty transcript + warning
A10	Truncated/corrupt MP4 (negative)	n/a	-	-	-	Graceful error, no crash

2.4 Upload file cases

ID	File name	Description	Expected result
F01	lesson_wiring.docx	Valid DOCX, 3 pages, 2 headings, 1 list	Accept
F02	manual_brakes.pdf	Valid text PDF, 10 pages	Accept
F03	notes.txt	UTF-8 text, 40 lines	Accept
F04	demo_torque.mp3	Valid MP3, 2 min	Accept
F05	demo_drip.mp4	Valid MP4, 5 min	Accept
F06	scan_only.pdf	Image-only PDF (no text layer)	Accept with 'no extractable text' warning
F07	payload.exe.pdf	Executable renamed to .pdf (MIME mismatch)	Reject (400)
F08	empty.docx	0 bytes	Reject (400)
F09	../../etc/passwd.txt	Path-traversal filename	Sanitize name or reject
F10	max_size_plus_1.mp4	Configured max size + 1 byte	Reject (413)
F11	max_size_exact.mp4	Exactly configured max size	Accept
F12	clip.avi	Valid AVI (listed in paper)	Accept

3. Test cases

Priority: High = safety, privacy or core pipeline. Type: Positive, Negative, Boundary, Edge, Safety, Security, Privacy, Accessibility, Non-functional. Test IDs are prefixed by module: UPL upload, ASR, NLP, RET retrieval, TRN translation, LOC localization, TTS, EVAL, API, UI, PERF, PRIV.

ID	Module	Title	Input / data	Expected result	Pri.	Type
UPL-01	Upload	Upload valid DOCX	F01 + target=Hindi	202 Accepted, job id returned, status = queued	High	Positive
UPL-02	Upload	Text PDF keeps structure	F02	Headings and list items preserved in extracted text	High	Positive
UPL-03	Upload	Audio upload	F04	Job created; routed to ASR stage	High	Positive
UPL-04	Upload	Video upload	F05	Audio track extracted; routed to ASR	High	Positive
UPL-05	Upload	Spoofed extension	F07	Rejected with 400; file not stored	High	Negative
UPL-06	Upload	Zero-byte file	F08	Rejected with clear message	Medium	Negative
UPL-07	Upload	Size boundary	F10 and F11	F11 accepted; F10 rejected with 413	Medium	Boundary
UPL-08	Upload	Image-only PDF	F06	Job completes with 'no text extracted' warning	Medium	Edge
UPL-09	Upload	Path-traversal filename	F09	Stored under a generated safe name; no file written outside upload dir	High	Security
ASR-01	ASR	Clean short audio	A01	WER <= 5%	High	Positive
ASR-02	ASR	Technical vocabulary	A05	WER <= 9.5%; G01-G16 terms recognized	High	Positive
ASR-03	ASR	Accent robustness	A06, A07	WER <= 11.5% each	Medium	Positive
ASR-04	ASR	Noisy workshop audio	A03	WER <= 10%; no timeout	Medium	Edge
ASR-05	ASR	Silence	A09	Empty transcript + warning; pipeline stops cleanly	Medium	Negative
ASR-06	ASR	Code-mixed speech	A08	Transcript produced; WER reported (no pass/fail bound)	Low	Edge
NLP-01	Preprocess	Sentence segmentation	S01-S16 joined as one paragraph	16 sentences recovered; no split inside 'N·m', '1:4', '121 °C'	High	Positive
NLP-02	Preprocess	Protected tokens	S16	'AX-4471', '230 V', '16 A', '10W-40' flagged do-not-translate	High	Positive
NLP-03	Preprocess	Empty / whitespace input	" and ' '	Handled with 400 / warning, no exception	Medium	Negative
RET-01	Retrieval	Exact term lookup	'fuse' (G01)	G01 returned as top-1 with its Hindi entry	High	Positive
RET-02	Retrieval	Inflected / paraphrased term	'blown fuses', 'drip-irrigation system'	G01 and G08 retrieved in top-3	High	Positive
RET-03	Retrieval	Out-of-glossary term	'oscilloscope' (not in sample glossary)	No constraint injected; segment flagged 'unmatched term'	Medium	Negative
RET-04	Retrieval	Domain disambiguation	S02 (electrical) vs 'earth the soil around the plant' (agri)	Electrical sense chosen for S02; no G02 constraint for agri sentence	High	Edge
RET-05	Retrieval	Glossary update without retraining	Add new entry, re-run S14	New rendering used; no model redeploy required	Medium	Positive
TRN-01	Translation	Negation preserved	S10, S04	Output contains negation ('do not') in each target language	High	Safety
TRN-02	Translation	Numbers and units	S03, S08, S09, S15	All numerals and units identical to source (e.g. 121 °C, 3 mm)	High	Safety
TRN-03	Translation	Glossary constraint honoured	S01 -> Hindi	Output contains the approved rendering of G01	High	Positive
TRN-04	Translation	Language smoke test	S01 x 22 scheduled languages	Non-empty output for every language; failures listed	Medium	Positive
TRN-05	Translation	Very long input	~3,000-word manual	Chunked automatically; output order preserved	Medium	Boundary

ID	Module	Title	Input / data	Expected result	Pri.	Type
TRN-06	Translation	Unsupported language code	target='xx'	400 with list of supported codes	Medium	Negative
TRN-07	Translation	Translation-memory reuse	S01 submitted twice	Second run is a TM hit; identical output; lower latency	Low	Positive
LOC-01	Localization	Safety values untouched	S07, S09	Localization never alters dosage/temperature values	High	Safety
LOC-02	Localization	Unit policy	S07 (acre)	Behaviour follows the configured unit policy and is logged	Low	Edge
TTS-01	TTS	Audio for evaluated languages	S01 x 8 evaluated languages	Non-empty audio per language; duration within 0.5x-2x of expected	High	Positive
TTS-02	TTS	Numeric reading	S08, S13	Numbers/ratios spoken correctly (listener check, 3 reviewers)	Medium	Positive
TTS-03	TTS	Missing voice model	Language with model removed	Fallback to text output + warning; job not failed	Medium	Negative
EVAL-01	Evaluation	BLEU identity	hyp == ref	BLEU = 1.0	High	Positive
EVAL-02	Evaluation	Low-BLEU gating	Segment BLEU < 0.62	Segment flagged for human review; not auto-published	High	Positive
EVAL-03	Evaluation	WER formula	Worked example (Section 4)	WER matches hand calculation	High	Positive
EVAL-04	Evaluation	Terminology accuracy	9 of 10 terms correct	90%	Medium	Positive
EVAL-05	Evaluation	Empty hypothesis	hyp = ""	BLEU = 0, WER = 100%, no division error	Medium	Boundary
API-01	API	Stage events	Any job	Ten stage events emitted in the designed order	Medium	Positive
API-02	API	Unknown job id	GET /jobs/unknown	404 with JSON error body	Low	Negative
API-03	API	Concurrency	20 simultaneous jobs	All complete; no cross-job data leakage	Medium	Non-functional
API-04	API	Reuse of upstream artifacts	Same file, 2nd language	ASR/NLP skipped; only translation onward runs	Medium	Positive
UI-01	UI	Language selection and preview	Select Tamil then Hindi	Correct output shown per selection	Medium	Positive
UI-02	UI	Progress display	Long job	Progress reflects stage events; no stall at 99%	Low	Positive
UI-03	UI	In-browser audio	Web Speech API playback	Audio plays without download; fails gracefully if unsupported	Medium	Positive
UI-04	UI	Screen-reader support	NVDA/TalkBack walkthrough	All controls labelled; focus order logical	High	Accessibility
PERF-01	Performance	10-minute audio end-to-end	A02-class input	Completes in under 4 min on reference hardware (paper claim)	Medium	Non-functional
PERF-02	Performance	Stage timing	10-min audio	Per-stage % recorded; translation + ASR dominate	Low	Non-functional
PRIV-01	Privacy	Raw audio deletion	Retention window set to 1 h	Raw audio deleted after window unless instructor opts in	High	Privacy
PRIV-02	Privacy	Opt-in retention	Instructor opt-in	Audio retained and access restricted to that instructor	Medium	Privacy
PRIV-03	Privacy	Unauthenticated access	Request without token	401; no job data returned	High	Security

4. Worked metric examples (verified by script)

These oracles let you check that your WER, BLEU and terminology functions are implemented correctly (tests EVAL-01 to EVAL-05). BLEU uses 4-gram modified precision with brevity penalty as in Eq. (1)-(3) of the paper; a simple add-half smoothing is applied for zero n-gram matches at sentence level.

Example	Reference	Hypothesis	Result
WER	switch off the main supply before replacing the fuse	switch of the main supply before replacing the fuels	S=2, D=0, I=0, N=9, WER = 22.2%
WER, empty hypothesis	(same reference)	(empty)	D=9, N=9, WER = 100%
BLEU identical	replace karne se pehle mukhya aapoorti band karen aur fyooz badlen	identical	BLEU = 1.0
BLEU close	(same reference)	...band karen phir fyooz badlen (one word changed)	BLEU = 0.742
BLEU poor	(same reference)	fuze badalne se pehle main supply bandh	BLEU = 0.088 (below the 0.62 gate, so flagged)
BLEU empty	(same reference)	(empty)	BLEU = 0.0
Terminology accuracy	10 glossary terms in test passage	9 translated correctly	Term. Acc. = 9/10 x 100 = 90%

The BLEU examples use Roman-script Hindi for illustration; they are not evidence about IndicTrans2 quality.

5. Originality check

What was checked. All 16 sentences, 16 glossary rows (term plus error note) and 53 test cases (title, input and expected text) were tested three ways. Word n-grams were lower-cased and stripped of punctuation.

Check	Result	Verdict
Overlap vs IEEE paper	0 of 722 4-grams; 0 of 637 5-grams	Pass: no shared 4-word phrase
Overlap vs IDL-I deck	0 of 722 4-grams; 0 of 637 5-grams	Pass: no shared 4-word phrase
Longest shared word run, any item vs paper	3 words (UPL-02)	Pass (generic words such as 'upload' and 'format')
Longest shared word run, any item vs deck	3 words (UI-04)	Pass
Duplicates inside the pack	Duplicate sentences: 0; duplicate test IDs: 0; duplicate test titles: 0	Pass
Near-duplicates inside the pack	Highest word-overlap (Jaccard): sentences 0.15 (S09/S11); tests 0.25 (TRN-04/TTS-01)	Pass (all below 0.30)
Cross-reference integrity	Every glossary ID cited in a sentence exists, and its term appears in that sentence	Pass

What this does not prove. I have no access to Turnitin, iThenticate or web-wide plagiarism databases from this environment, so I could not scan the open web or published test suites. The check shows the pack does not reuse wording from your own two documents and contains no internal repetition. Short operational sentences such as 'switch off the main supply' are normal phrases in electrical safety manuals, so a web scan may show small coincidental matches. Before submission, run the final report through your college's similarity tool.

Disclosure. This data and these test cases were generated by an AI assistant. Many institutions require AI assistance to be declared, so check the IDL-I guidelines and add an acknowledgement if needed. Hindi/Tamil renderings and the gold-standard reference translations must come from your human experts, which also makes that part of the dataset yours.

6. Inconsistencies in the supplied documents that the tests should catch

While building the oracles I cross-read the paper and deck. These are worth fixing or explaining, because a reviewer who recomputes numbers will find them.

Topic	Observation
Survey size	Figs. 2-6 percentages (7.1%, 14.3%, 21.4%, 28.6%, 35.7%, 42.9%, 57.1%, 64.3%, 71.4%, 85.7%) are all multiples of 1/14, implying about 14 respondents, while the abstract and conclusion cite 30 learners for the 87% audio-clarity result. State both samples explicitly.
BLEU range	Abstract says 0.62-0.78; Table III and Section X give 0.68-0.78. 0.62 is the deployment threshold, not a measured score.
Average BLEU	Table III mean is 0.72 (5.78/8); Table V reports 0.73 for full VoiceVerse. Terminology mean (87.75%) matches the 88% in Table V.
Headline 91%	91% is the Tamil figure in Table III; the mean across the eight languages is about 88%. Say which one the 91% vs 63% comparison uses.
Language coverage	Abstract claims 22 languages; the evaluation covers 8. Deck mentions VR/3D and 'global languages', which the paper does not implement.
Section references	Text refers to 'Section X' and 'Section IX' for fine-tuning and WER gaps, but those discussions are in Sections VII-C and XI. 'Section VI-B/C' citations in Section VI also point to Section VII results.
Deck vs paper	The deck's interview slide says no formal interviews were done; the paper's survey and user study should be described as such, not as validation.